

BU5300 – IT Project Management

Lecture 1: Overview, traditional and agile approaches

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Acknowledgement

Large part of the contents of this course
were originally developed by
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About me

Jefferson Seide Molléri

- PhD in Software Engineering (focus on Empirical Research Quality)
- PostDoc in IT Management
- 10+ years of industrial experience, many of which as project manager on software development w/agile
- **Other interests:** software process, process improvement, agile, user interface, gamification
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About me

Daniel Isdal Furset

- MSc in Innovation & Management
- 5+ years of industrial experience, many of which as project manager on various projects.
- Currently work as the Secretary General of a Norwegian industry association called Fintech Norway. An association by and for FinTech companies operating in the Norwegian market. I also have a board position at EDFA – European Digital Finance Association – focusing mainly on lobbying towards the EU on behalf of the FinTech industry.
- I have previously lectured in Digital Business Understanding (Digital Forretningsforståelse), in addition to a number of other courses on innovation, management, and organisational subjects, at both Høgskolen Kristiania and Høgskolen på Vestlandet (HVL) in Bergen.
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Agenda for today's lecture

- What is project management?
- Traditional vs Agile project management
- Overview of the course
- Types of Information Systems Projects (Chapter 1)
- Business strategy and information systems (Chapter 2)

What is project management?

A definition and overview

What is a Project?

by the Project Management Institute (PMI)

- “A temporary group activity designed to produce a unique product, service, or result”.
- Project characteristics
 - has a start and an end
 - has an organization and steering committee
 - has a clear mandate and goals for what to produce – often referred to as ‘deliverables’
 - has a defined plan of activities, budget, and schedule

What is Project Management?

by the Association of Project Management (UK)



<https://www.youtube.com/watch?v=Jk-JwtScIIw>

What is Project Management?

by the Project Management Institute (PMI)

- Project management is the use of specific knowledge, skills, tools and techniques to deliver something of **value** to people.
- The **development of software for an improved business process**, the construction of a building, the relief effort after a natural disaster, the expansion of sales into a new geographic market — these are all examples of projects.

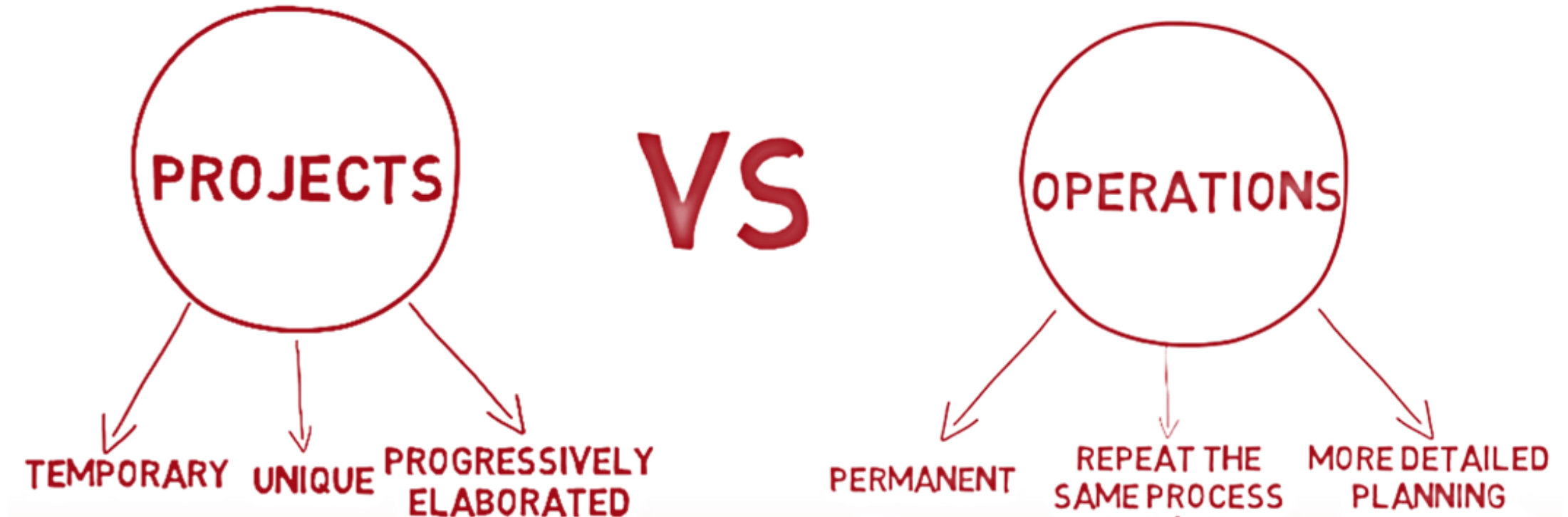
<https://www.pmi.org/about/learn-about-pmi/what-is-project-management>

What is a Project Manager?

by the Project Management Institute (PMI)

- The project Manager is responsible for managing the project in terms of plans, budgets, resources, competencies, and communication with relevant actors (e.g., steering committee, functional manager)
- **Basic assumption:** If correctly planned and executed, a project will achieve its goals (deliver its value).

The two modes of work in business organizations

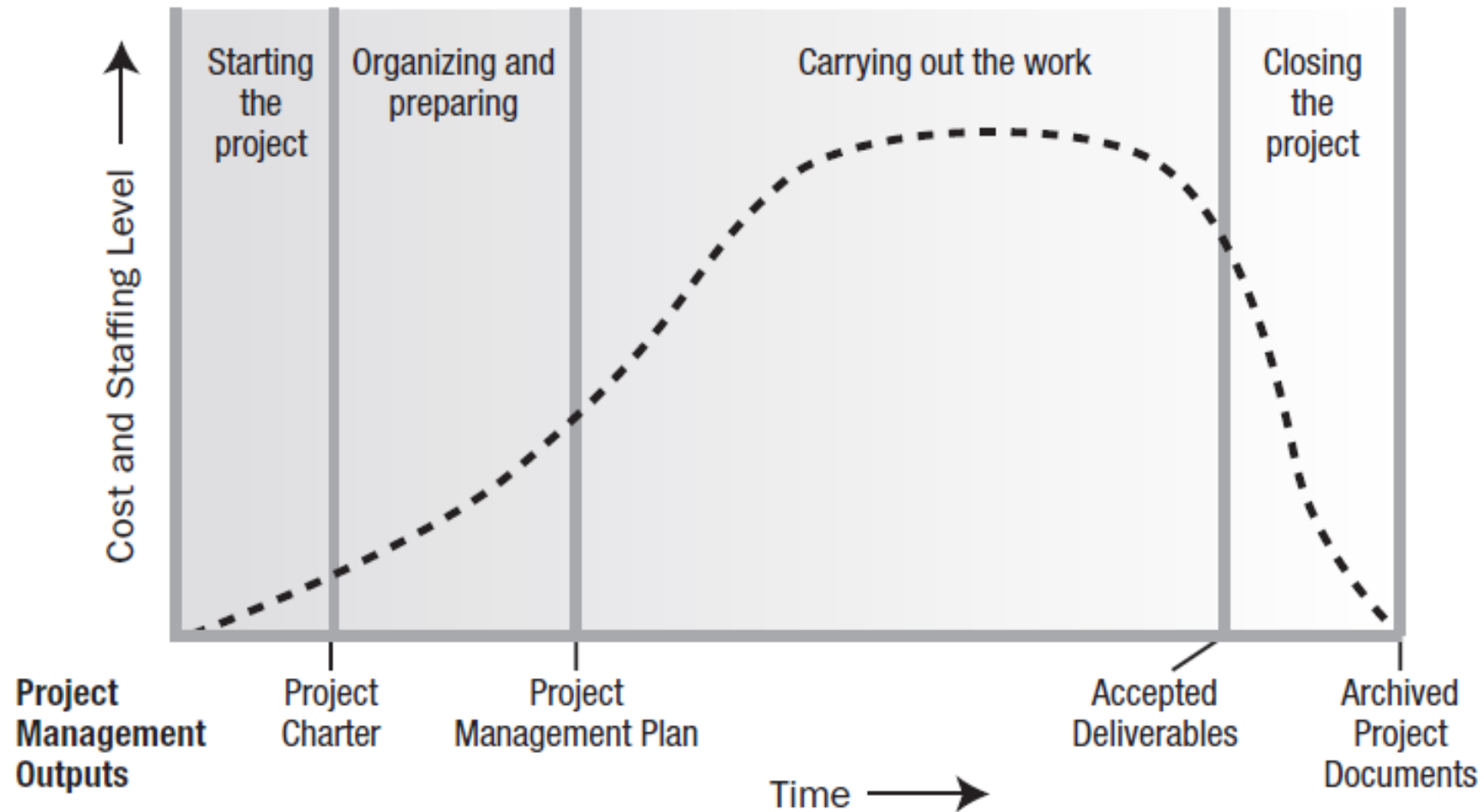


Why organize activities as a project?

- Solve complex tasks that the line organization is not designed to do
 - Example: develop a complex information system at hospitals
 - Example: develop a case-handling information system at NAV (Norwegian welfare organization)
- Hire a consulting company in order to have access to appropriate knowledge and competence
- Minimize risk for the organization

Project life cycle

by the Project Management Institute (PMI)

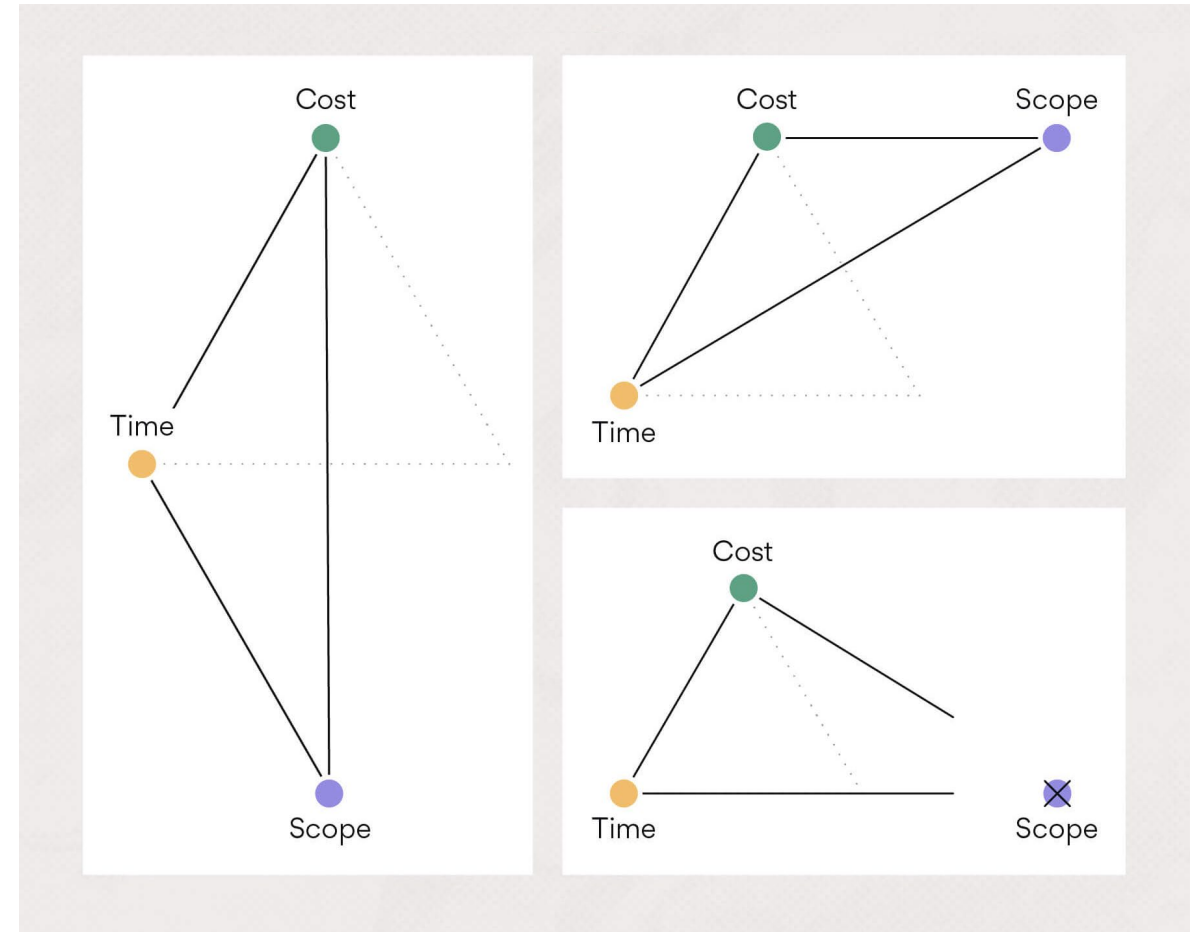


Project life cycle breakdown



Time / cost / scope triangle or “working with triple constraints”

- Project management must balance scope, time, and cost
- Changing one aspect has consequences for the others



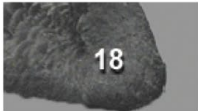
20% of all IT projects are "Black Swans"

- On average, IT projects in public sector perform remarkably well!
- But, large-scale projects are especially prone to risks and failures
- Projects that implicate standard software are challenging!!!

Flyvbjerg, Bent, and Alexander Budzier, **Why Your IT Project May Be Riskier Than You Think**, Harvard Business Review. Vol. 89 (2011), No. 9, pp. 23-25.

Black swans have high cost, schedule and benefit risks, which are hidden in the fat tails of ICT portfolios

Risk comparison, Median, in percent

	Black Swans ¹	Projects with cost overrun	Normal projects	Starved projects
Cost overrun	 +130	+47	+0	-75
Schedule overrun	 +41	+38	+24	n/a
Likelihood	 18	28	31	51

- Even if the median cost overrun is low, risk of cost overruns is high
- **Black Swans mean very high cost and schedule risks**
- And all the projects with a downside risk show significant risk

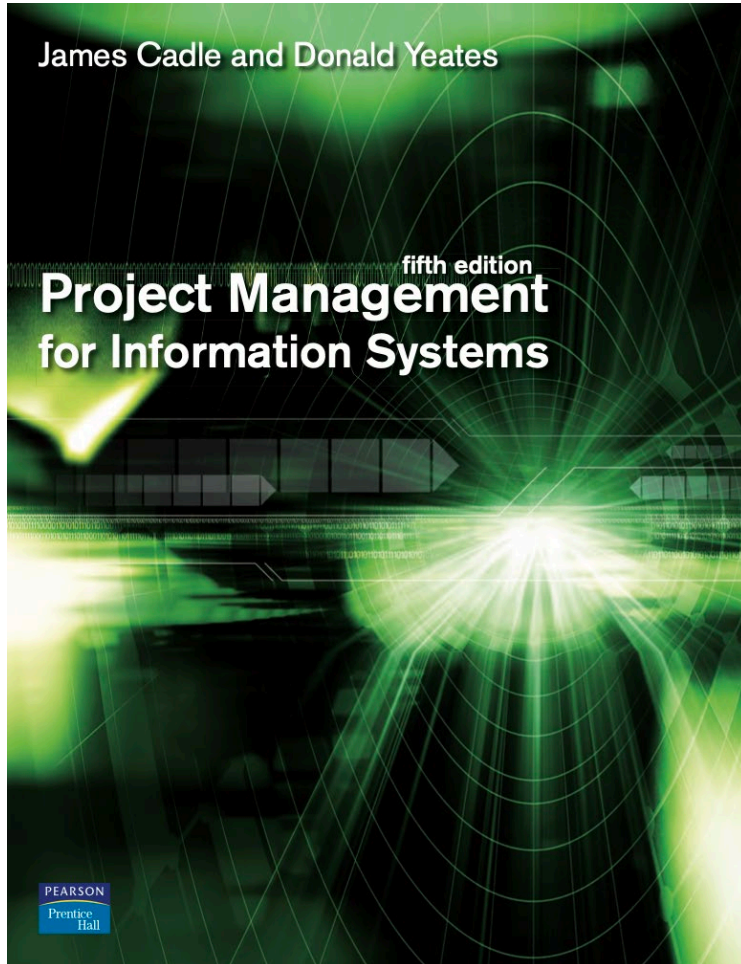
¹ The statistical expectation value does not converge and is infinite

Overview of the course

Overview of the course

- Lectures weekly
 - The schedule is on TimeEdit; but changes can occur > check Canvas often
 - Guest lectures on specific topics
- All lecture slides will be shared on Canvas
 - Also, the textbook is available at the Library
 - Articles that elaborate on the topic in the textbook (but not replace)
- Course work
 - Exercises at the end of each lecture
 - Case-based learning (group work)

Textbook (Pensum)



The following chapters are the curriculum:

- 1 – Types of information systems Projects
- 2 – Business strategy and information systems
- 3 – The business case
- 4 – The organizational framework
- 6 – Development lifecycles and approaches
- 7 – The profile of a project
- 8 – Project planning: understanding the work
- 9 – Project planning: estimating
- 10 – Project planning: scheduling and resourcing
- 11 – Monitoring progress
- 12 – Exercising control
- 13 – Reporting progress
- 14 – Managing quality
- 15 – Managing risk
- 17 – Selling the project
- 18 – Managing stakeholders
- 20 – Managing change
- 21 – Leadership and performance
- 22 – Managing the team
- 23 – The project manager

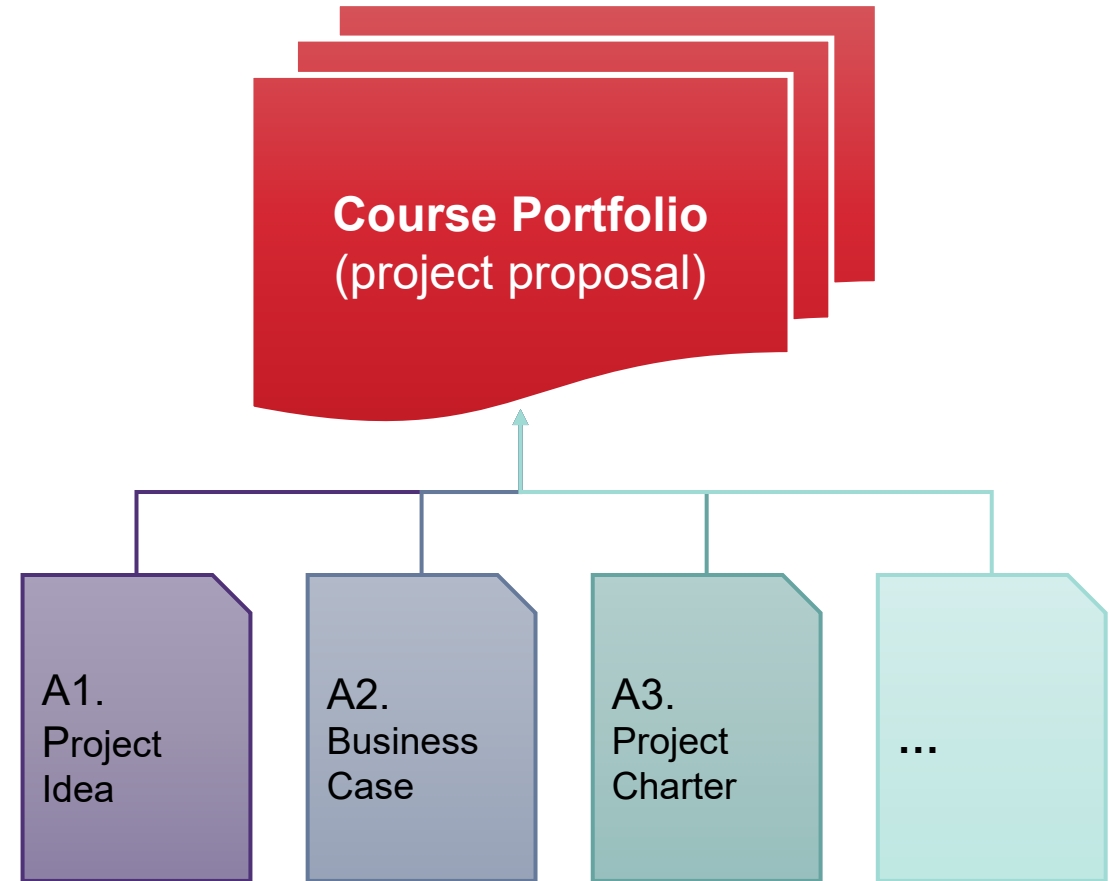
Articles included in the curriculum

- Elaborate on topics from the textbook with different cases, theory, and examples
- The articles provide insight to reflect on IT project management and different situations one may face
- Not expected to remember all the details from the articles – but the main points of argumentation, theory and cases

Weekly Course Activities

Group Work

- Students collaborate in *fixed* groups throughout the semester
- Each lecture gradually develops a project idea into a full IT project proposal
- Feedback provided via Canvas
- Every activities address different components of the course portfolio (final exam)



Course Activity 1

Starting today!

- Form a group (6–8 students) and create a Project Group in Canvas
- Brainstorm an **IT Project Idea**
 - Check the case example in Canvas: *Smart Tourist Navigation for VisitOslo*
- Detail the following:
 - A summary of the project (1-2 paragraphs)
 - The type of the project (which category of the 9 nine categories)
 - Potential strengths, weaknesses, opportunities and threats (SWOT analysis)
 - What type of technical skills are needed for such a project
- **Note!** In the next lecture, we'll walk you through how to define the problem, check that it actually exists, and validate your idea
- Submit it to Canvas: Assignment 1 (**check deadline**)

Examination

Individual submission of a group work?

- Structure
 - Group Portfolio (8–10 pages)
 - Individual Reflection (1 page)
- Assessment Criteria
 - Completeness
 - Professional presentation
 - Self-Reflection and participation

Important!

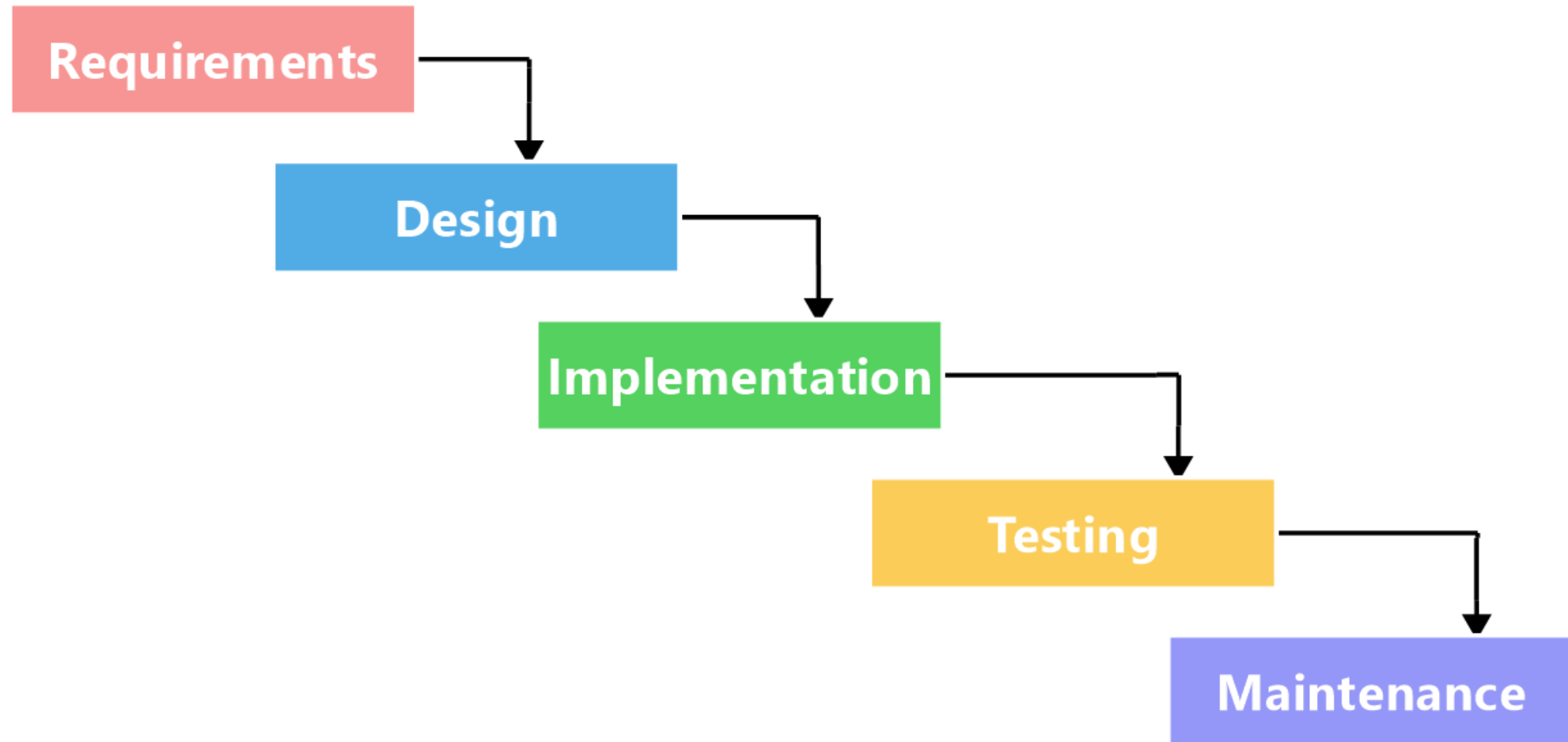
- The final proposal should be built from the weekly assignments
- Grounded on theory and concepts from the curriculum
- Feedback provided during the semester should be incorporated

Traditional vs Agile project management

Traditional approaches

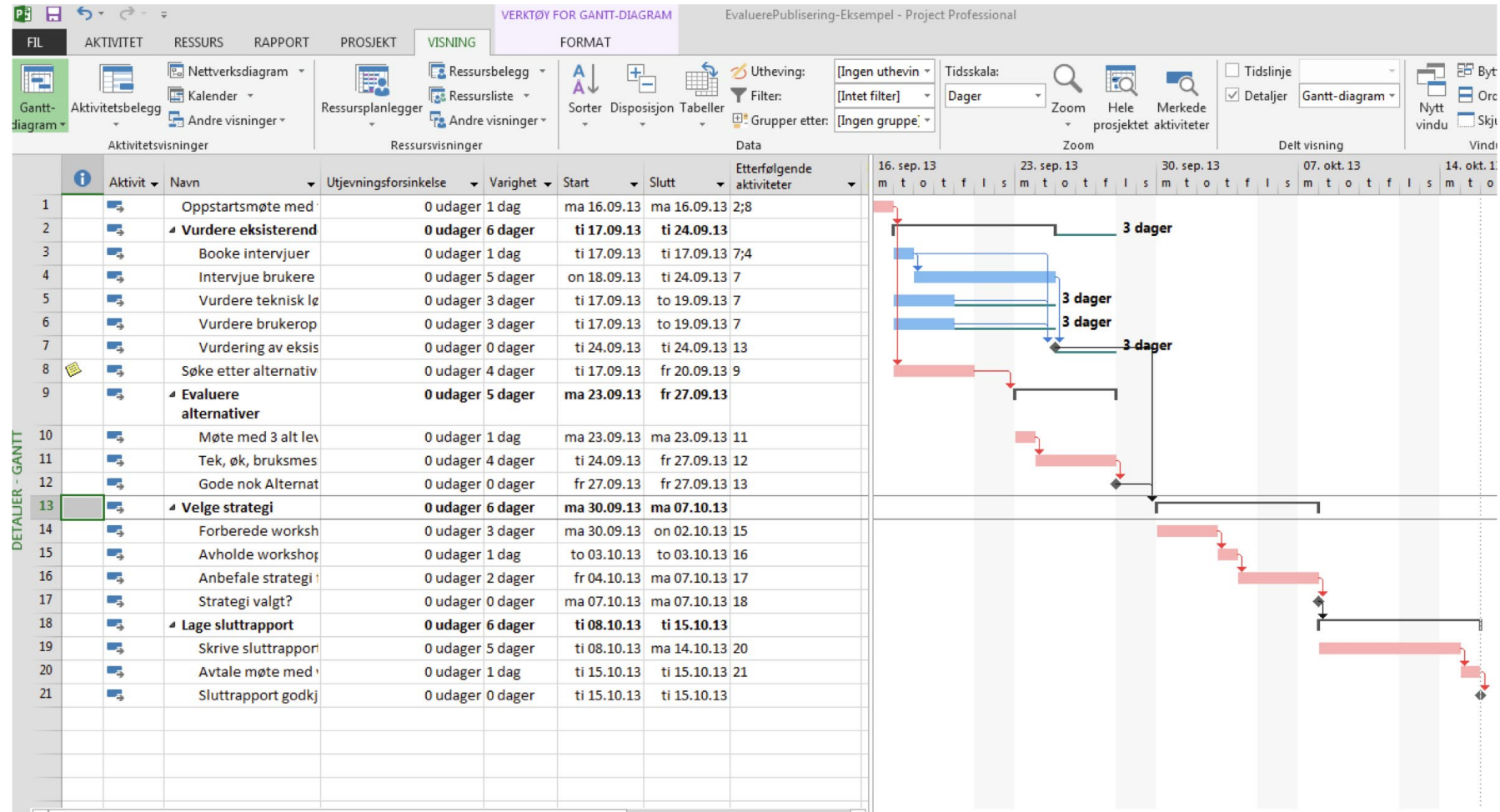
- Focus on management control and planning
- Detailed plans carried out in a waterfall-like fashion
- Hierarchical organizations often with a top- down approach to project governance
- IT-projects: requirements must be defined up-front and not change too much

Waterfall methodology



GANTT chart

A tool for managing traditional projects



Agile project management approaches

- Co-located and small teams
- Coordination through physical artifacts as well as software tools
- Close interaction with customer
- Self-managed teams

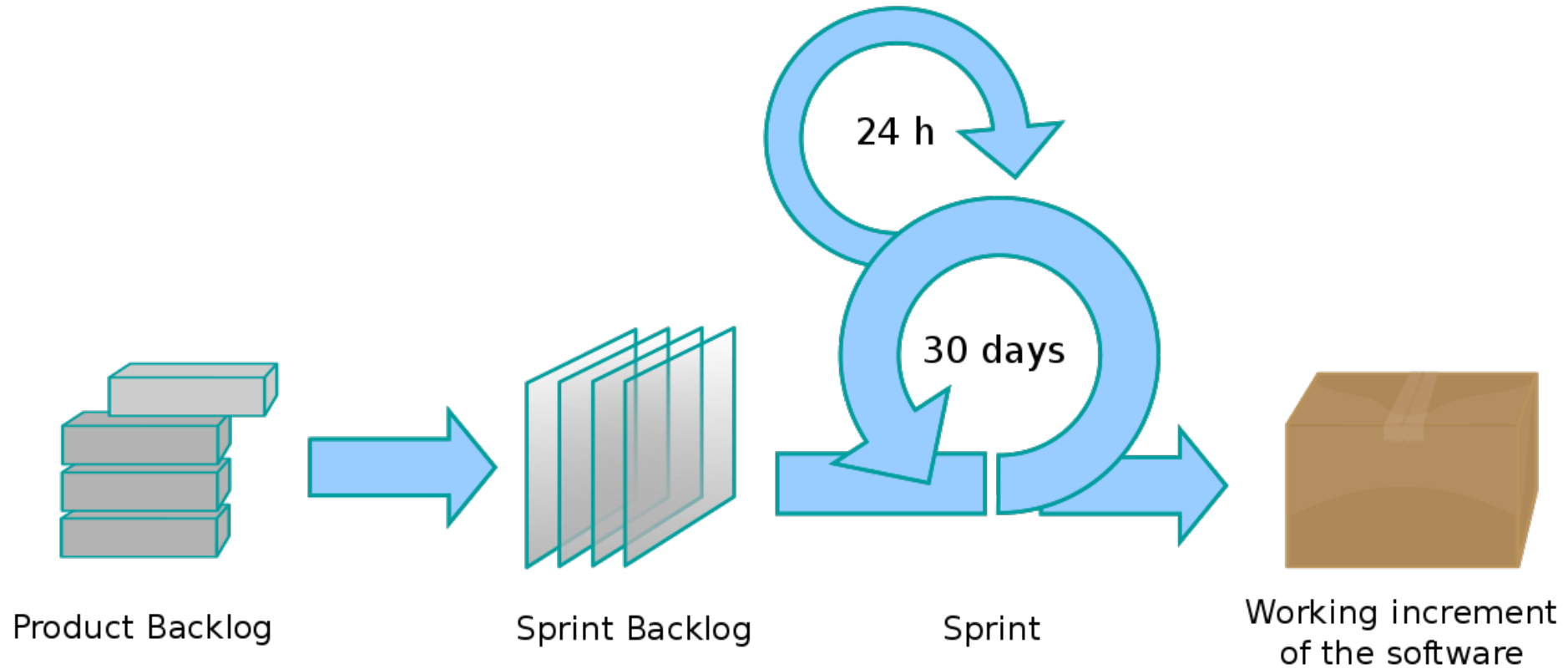
Final Definition of Agility

the continual readiness of an ISD method to rapidly or inherently create change, proactively or reactively embrace change, and learn from change while contributing to perceived customer value (economy, quality, and simplicity), through its collective components and relationships with its environment.

Conboy, Kieran. Agility from first principles: **Reconstructing the concept of agility in information systems development**. Information systems research 20.3 (2009): 329-354.

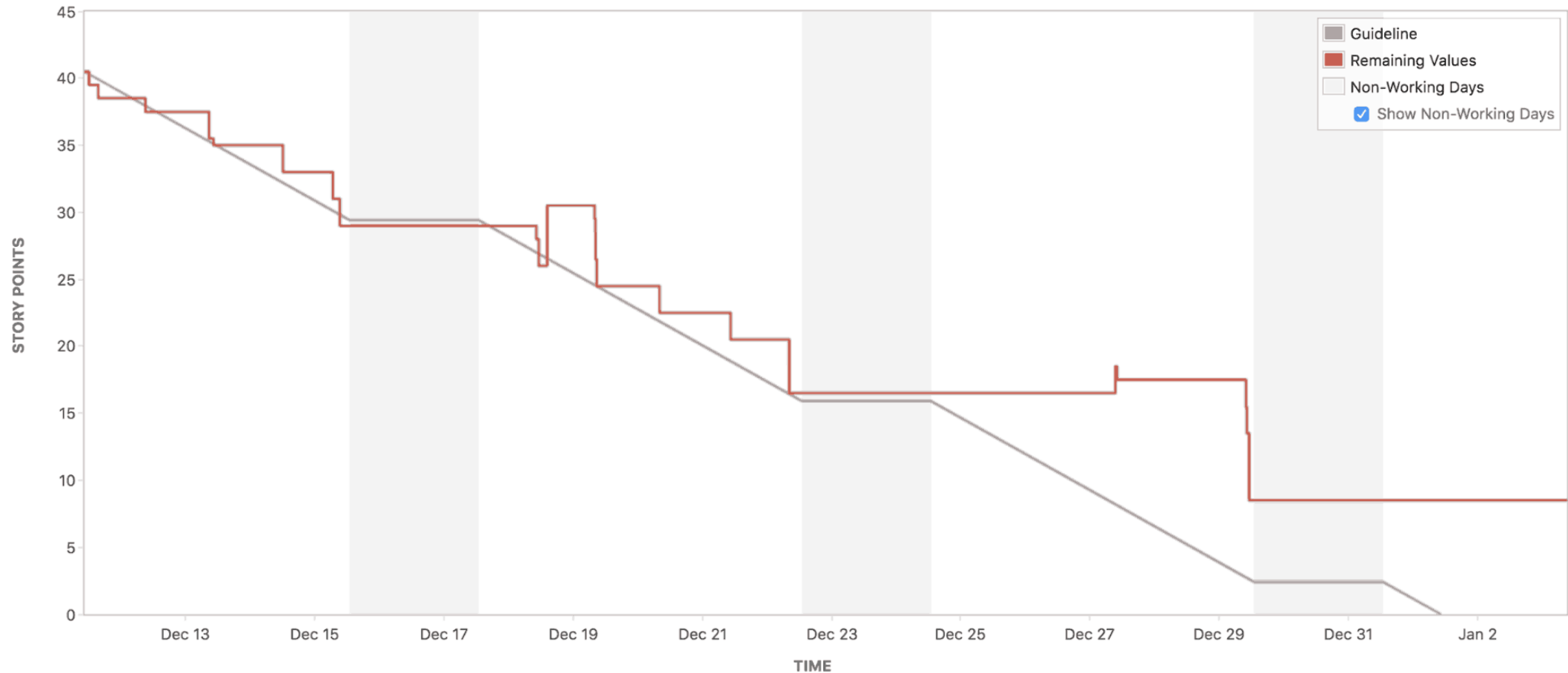
Scrum

A methodology for agile processes



Burn-down charts

A tool for managing sprints



<https://www.atlassian.com/agile/tutorials/burndown-charts>

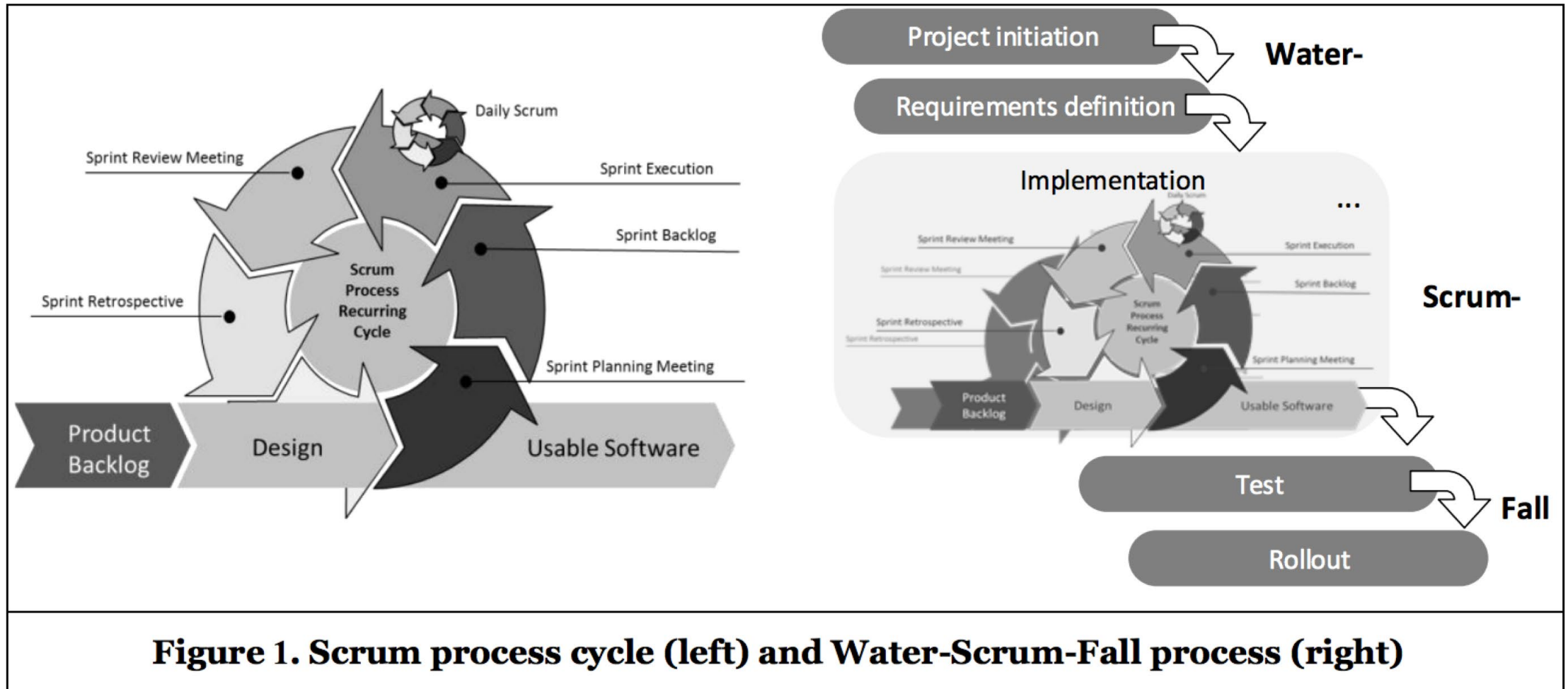
Traditional vs Agile

	Traditional	Agile
Fundamental Assumptions	Systems are fully specifiable, predictable, and can be built through meticulous and extensive planning.	High-quality, adaptive software can be developed by small teams using the principles of continuous design improvement and testing based on rapid feedback and change.
Control	Process centric	People centric
Management Style	Command-and-control	Leadership-and-collaboration
Knowledge Management	Explicit	Tacit
Role Assignment	Individual—favors specialization	Self-organizing teams—encourages role interchangeability
Communication	Formal	Informal
Customer's Role	Important	Critical
Project Cycle	Guided by tasks or activities	Guided by product features
Development Model	Life cycle model (Waterfall, Spiral, or some variation)	The evolutionary-delivery model
Desired Organizational Form/Structure	Mechanistic (bureaucratic with high formalization)	Organic (flexible and participative encouraging cooperative social action)
Technology	No restriction	Favors object-oriented technology

Nerur, Sridhar, RadhaKanta Mahapatra, and George Mangalaraj. **Challenges of migrating to agile methodologies.** *Communications of the ACM* 48.5 (2005): 72-78.

Mixed approaches

e.g. Water-Scrum-Fall



Schlauderer, Sebastian, Sven Overhage, and Björn Fehrenbach. **Widely used but also highly valued? Acceptance factors and their perceptions in water-scrum-fall projects.** Proceedings of the 36th International Conference on Information Systems (ICIS), Fort Worth, TX, USA 2015. AIS Electronic Library (AISeL), 2015.

Chapter 1:

Types of Information Systems Projects

Types of information systems projects

1. Software development
2. Package implementation
3. System enhancement
4. Consultancy and business analysis assignments
5. Systems migration
6. Infrastructure implementation
7. Outsourcing (and in-sourcing)
8. Disaster recovery
9. Smaller IS projects

Some IS projects embrace more than one of the nine categories

For example, a project may involve a package implementation, original software development and putting in a new infrastructure

Software development projects

- A group of people working together to specify, design, develop, test, and implement a new system for a 'customer' (either internal or external).
- Most of the traditional project management methods and techniques are applicable
 - Resembles construction projects but the product is intangible
- The emergence of 'agile' development methods is, in part, a response to the difficulty of specifying software requirements 'up front' and a recognition that a more evolutionary approach is needed
 - Flexibility of approach in being prepared to revisit the specification and negotiate with the customer as the project proceeds
 - Well-developed interpersonal and stakeholder-management skills

Package implementation projects

- In principle, package implementation is simple: the package is bought, installed, switched on and used
- Challenges:
 - Selecting the right package
 - Matching the package to the customer needs (configuration vs. customization)
 - Integration with other systems
- Requires significant “Organizational Change Management (OCM)” effort to ensure adoption and value realization.
- Requires skills in developing Request for Information/Proposal (RFI/RFP), contracting, and vendor management

Consultancy and business analysis assignments

- These projects are about investigating a business issue and proposing solutions using information technology
 - Example, how to improve customer retention
- May not involve developing or installing anything
- Challenges:
 - Customer may not be not quite sure what the problem is or where suitable solutions may be found
 - The scope of the project is difficult to estimate and define up front
 - Unclear scope requires flexible budget and timeline
- Contracts tend to be time and material (T&M) with an upper limit rather than fixed cost

Systems migration projects

- This type of project is one where an existing operational system must be moved to a new operating environment – perhaps because the current one is now longer supported or supportable
 - Examples: migrate from Windows 7 to Windows 11, migrate from Zoom to Panopto, etc
- May involve additional software development
- It may be necessary to create new interfaces with other systems
- There may also be infrastructure implications
- It might also be necessary to carry out some limited retraining of users to enable them to utilize the new environment
 - This is typically is part of organizational change management effort

Infrastructure projects

- For example, introduce or replace hardware, servers, or PCs; put in place communications infrastructures
- All traditional project management practices are applicable
- Usually need to maintain 'business as usual' whilst putting in place the new infrastructure
- Requires significant supplier management as most of the work is usually subcontracted and all the equipment is bought-in
- Managing firm and realistic timescales is necessary
- Understanding inter-dependency between tasks is also critical

Outsourcing (and in-sourcing) projects

- Reasons for outsourcing:
 - The wish to gain access to the expertise of the outsourcing providers
 - Difficulty in managing the IT estate internally
 - Reduce costs, through economies of scale or by taking advantage of lower labor costs
 - Commoditizing IT (Assuming it no longer provides a source of competitive advantage)
- Requires:
 - ‘due diligence’ in making sure the scope of the contract is clear and feasible
 - Training new people in the IT systems to be supported.
 - Establishing who will have ownership of assets and resources
 - Organizing the transfer of software licenses from one party to another
 - Managing layoffs and staff replacement
- Does it work? And how about Backsourcing?

Disaster recovery projects

- A key element for “Business Continuity”
- Involves the activation of already in place plans for recovering after a disaster
- Preventive defensive measures can reduce the occurrence of various threats
- Time is critical for these projects
- Pre-planning of the recovery process is by far the best way of ensuring successful recovery
- An excellent way of ensuring the success of DR processes is to carry out exercises to test them ‘drills’

Smaller IS projects

- Is it necessary to go through all the rigors of estimating, planning, monitoring, and reporting even for small projects?
- Project management should be pragmatic in nature
- Some steps should never be skipped like the “Project Initiation Document (PID)”
- Rough estimates of the scope and time needed may be enough
- Progress can be reported informally through emails

Other project classifications

- Internal projects: done within an organization
- External projects: carried out by another organization under contract

The textbook uses a conventional software development project to illustrate how the project management principles and techniques are applied

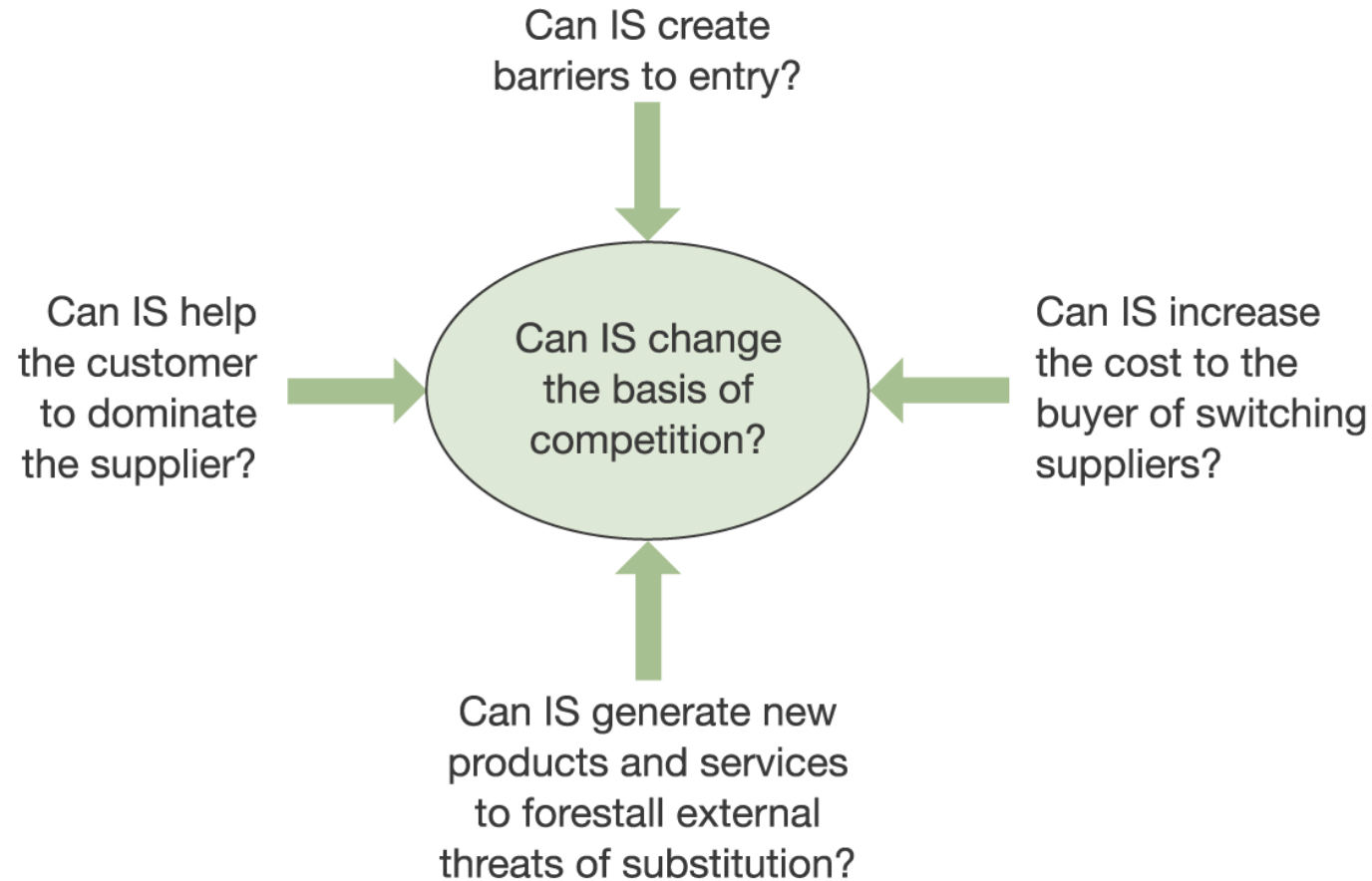
Chapter 2:

Business strategy and information systems

Business Strategy

- A business strategy is the backbone of every business
 - It comprises all the decisions taken, and actions undertaken for achieving a business vision
 - Knowing how to execute it properly can help businesses become successful
- It is vital that the information system aligns with the business strategy
- IS project ideas that does not align (or are unfeasible) are likely rejected

Robson's analysis of IS opportunities



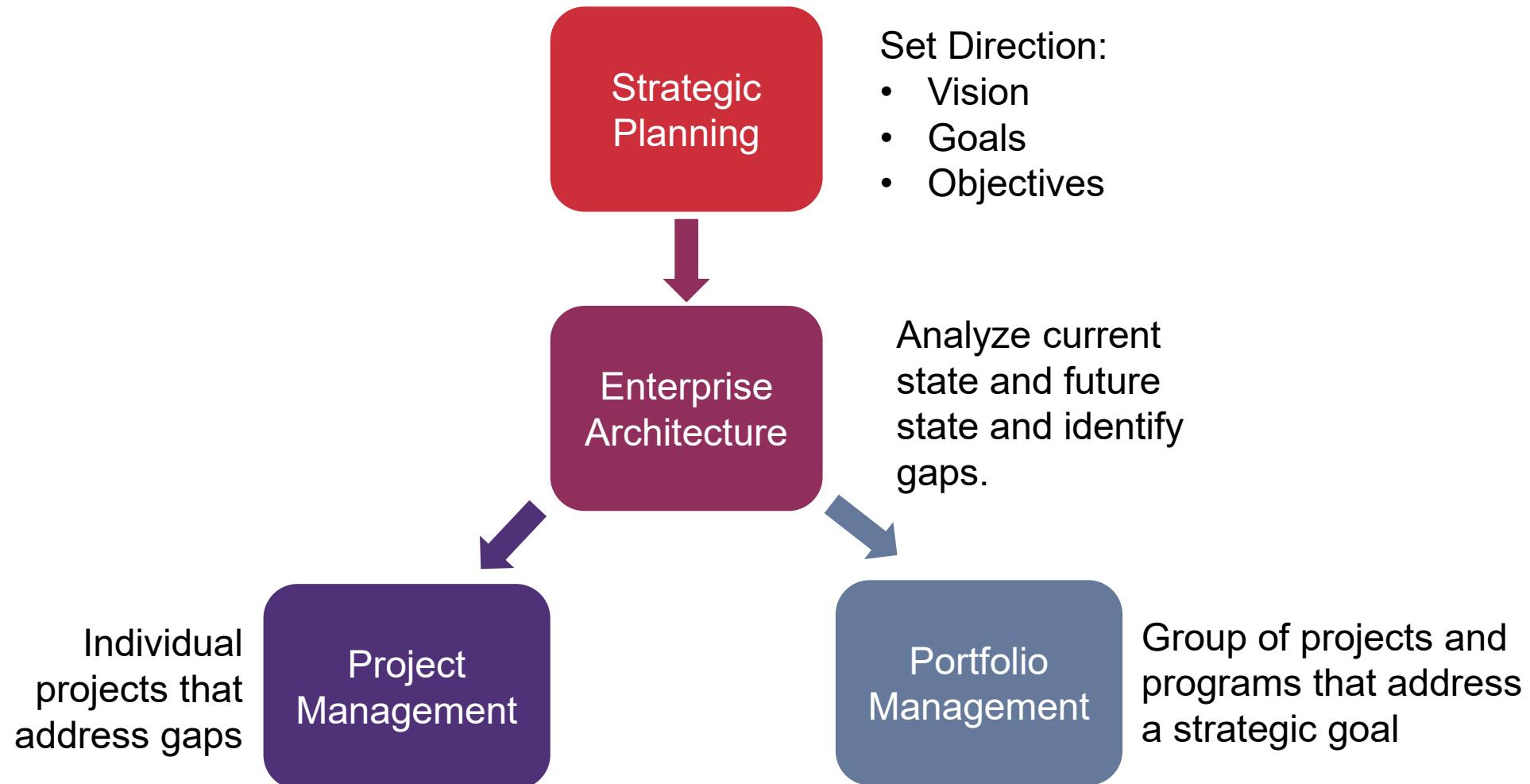
Boddy, David, Albert Boonstra, and Graham Kennedy. **Managing information systems: an organisational perspective.** Pearson Education, 2005.

SWOT Analysis



- **Strengths** describe what an organization excels at and what separates it from the competition
- **Weaknesses** stop an organization from performing at its optimum level
- **Opportunities** refer to favorable external factors that could give an organization a competitive advantage
- **Threats** refer to external factors that have the potential to harm an organization

Strategic Planning and Project Management



Questions?

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Bibliography

- Cadle, James, and Donald Yeates, eds. **Project management for information systems**. Fifth edition. Pearson education, 2008.
- Flyvbjerg, Bent, and Alexander Budzier, **Why Your IT Project May Be Riskier Than You Think**, Harvard Business Review. Vol. 89 (2011), no. 9, pp. 23-25.
- Špundak, Mario. **Mixed agile/traditional project management methodology–reality or illusion?**. Procedia-Social and Behavioral Sciences 119 (2014): 939-948.
- Conboy, Kieran. Agility from first principles: **Reconstructing the concept of agility in information systems development**. Information systems research 20.3 (2009): 329-354.
- Nerur, Sridhar, RadhaKanta Mahapatra, and George Mangalaraj. **Challenges of migrating to agile methodologies**. *Communications of the ACM* 48.5 (2005): 72-78.
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